La Scuola d'Italia / Dr. Huson / IB Math 11
14 October 2026

Nickname:

1.12 Homework: Review of Sequences and Series

1. Consider the arithmetic sequence

$$7,\ 11,\ 15,\ 19,\ \ldots$$

   (a) Write down the common difference d.

   (b) Find an expression for u_n, the nth term.

   (c) Find u_{25}.

   (d) Find which term of the sequence is equal to 143.

2. In an arithmetic sequence, $u_3 = 14$ and $u_8 = 39$.

   (a) Find the common difference d and the first term u_1.

   (b) Find u_{20}.

   (c) Find S_{20}, the sum of the first 20 terms.

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3. Consider the arithmetic series

$$5 + 9 + 13 + 17 + \cdots + 201.$$

(a) Find the number of terms in the series.

(b) Find the sum of the series.

4. A theatre has 25 rows of seats. There are 20 seats in the first row, and each row has 2 more seats than the row in front of it.

(a) Find the number of seats in the last row.

(b) Find the total number of seats in the theatre.

(c) A school group needs at least 600 seats. The group fills the rows in order, starting from the first row. Find the least number of rows the group needs.

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5. Consider the geometric sequence

$$3,\ 6,\ 12,\ 24,\ \ldots$$

(a) Write down the common ratio r.

(b) Find an expression for u_n, the nth term.

(c) Find u_{10}.

(d) Find the first term of the sequence that is greater than 10 000, and state which term it is.

6. In a geometric sequence, $u_2 = 12$ and $u_5 = 324$.

(a) Find the common ratio r and the first term u_1.

(b) Find u_7.

(c) Find S_6, the sum of the first six terms.

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7. Giulia is training for a race. She runs 5 km in the first week, and each week she increases her distance by 10%.

   (a) Write down the common ratio of the weekly distances.

   (b) Find the distance she runs in the eighth week.

   (c) Find the total distance she runs in the first eight weeks.

8. Marco invests €5000 in a savings account that pays 4% per year, compounded **quarterly**.

   (a) Find the value of the investment after 6 years.

   (b) Find the interest Marco earns in these 6 years.

   (c) Another account pays 4% per year, compounded **annually**. Find the value of €5000 invested in this account after 6 years.

   (d) For the quarterly account, find the number of complete years after which the value of Marco's investment first exceeds €8000.